WORKSHEET

Chapter 8: Uniform and triangular probability distribution functions

1 Find the uniform probability density functions for continuous random variables defined on the following domains.

a [0, 10]

b [1, 15]

c [4, 12]

d [8, 28]

2 If the continuous random variable, X, has a uniform probability density function on the domain [2, 26], calculate the following probabilities.

a $x \geq 12$

b $x \leq 10$

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c $3 \leq x \leq 9$

d $4 \leq x \leq 20$

3 Find the triangular probability density functions with maximum values at the centres for continuous random variables defined on the following domains.

a [0, 10]

b [2, 14]

c [5, 15]

d [4, 28]

4 Find the triangular probability density functions for continuous random variables defined on the following domains.

a [5, 20] with a maximum value at 7

b [2, 18] with a maximum value at 12

c [1, 21] with a maximum value at 15

d [8, 32] with a maximum value at 14

5 A fast-food store is planning a new outlet and uses a triangular distribution to model the future weekly sales. It is estimated that the minimum weekly sales is $1000 and the maximum is $6000. It is also estimated that the most likely outcome is around $3000. The new outlet needs to achieve weekly sales of $2000 to cover its costs.

What is the probability that the outlet won't cover costs in a week?

6 A candidate in a local council election thinks the most likely number of votes she will receive is around 550, but she could get as many as 900 or as few as 200. What is the probability she will receive more than 450 votes?

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Answers

1 **a** $p(x) = \begin{cases} \frac{1}{10} & \text{for } 0 \leq x \leq 10 \\ 0 & \text{for } x < 0 \text{ or } x > 10 \end{cases}$

b $p(x) = \begin{cases} \frac{1}{15} & \text{for } 1 \leq x \leq 14 \\ 0 & \text{for } x < 1 \text{ or } x > 14 \end{cases}$

c $p(x) = \begin{cases} \frac{1}{8} & \text{for } 4 \leq x \leq 8 \\ 0 & \text{for } x < 4 \text{ or } x > 8 \end{cases}$

d $p(x) = \begin{cases} \frac{1}{20} & \text{for } 8 \leq x \leq 28 \\ 0 & \text{for } x < 8 \text{ or } x > 28 \end{cases}$

2 **a** $\frac{1}{2}$

b $\frac{5}{12}$

c $\frac{1}{4}$

d $\frac{2}{3}$

3 **a** $p(x) = \begin{cases} \frac{1}{25}x & \text{for } 0 \leq x < 5 \\ -\frac{1}{25}(x - 10) & \text{for } 5 \leq x \leq 10 \end{cases}$

b $p(x) = \begin{cases} \frac{1}{36}(x - 2) & \text{for } 2 \leq x < 8 \\ -\frac{1}{36}(x - 14) & \text{for } 8 \leq x \leq 14 \end{cases}$

c $p(x) = \begin{cases} \frac{1}{25}(x - 5) & \text{for } 5 \leq x < 10 \\ -\frac{1}{25}(x - 15) & \text{for } 10 \leq x \leq 15 \end{cases}$

d $p(x) = \begin{cases} \frac{1}{144}(x - 4) & \text{for } 4 \leq x < 16 \\ -\frac{1}{144}(x - 28) & \text{for } 16 \leq x \leq 28 \end{cases}$

4 **a** $p(x) = \begin{cases} \frac{1}{15}(x - 5) & \text{for } 5 \leq x < 7 \\ -\frac{2}{195}(x - 20) & \text{for } 7 \leq x \leq 20 \end{cases}$

b $p(x) = \begin{cases} \frac{1}{80}(x - 2) & \text{for } 2 \leq x < 12 \\ -\frac{1}{48}(x - 18) & \text{for } 12 \leq x \leq 18 \end{cases}$

c $p(x) = \begin{cases} \frac{1}{140}(x - 1) & \text{for } 1 \leq x < 15 \\ -\frac{1}{60}(x - 21) & \text{for } 15 \leq x \leq 21 \end{cases}$

d $p(x) = \begin{cases} \frac{1}{172}(x - 8) & \text{for } 8 \leq x < 14 \\ -\frac{1}{96}(x - 32) & \text{for } 14 \leq x \leq 32 \end{cases}$

5 0.1

6 0.7449

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